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Liquid ejection head and liquid ejection apparatus

Abstract

A liquid ejection apparatus includes a printing element board, supply and collection flow passages, a liquid feeding mechanism, and first and second bubble reservoir units. The printing element board includes a pressure chamber having an ejection port from which the printing element board ejects a liquid. The supply and collection flow passages communicate with the pressure chamber. The liquid feeding mechanism generates a difference in pressure between the supply and collection flow passages to supply the liquid from the supply flow passage to the pressure chamber and to recover the liquid in the pressure chamber from the collection flow passage. The first bubble reservoir unit connects the supply flow passage to the liquid feeding mechanism. The second bubble reservoir unit connects the collection flow passage to the liquid feeding mechanism. A volume of the first bubble reservoir unit is larger than a volume of the second bubble reservoir unit.

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Background/Summary

BACKGROUND
Field
(1) The present disclosure relates to a liquid ejection head and a liquid ejection apparatus.
Description of the Related Art
(2) Japanese Patent Laid-Open No. 2003-312006 discloses a liquid ejection head that which includes a fluid reservoir, a pump, a circulation flow passage, and a print head, which are provided to a carriage, and is configured to circulate a fluid into the circulation flow passage with the pump and to supply the fluid from the fluid reservoir to the print head during a printing cycle.
(3) However, the liquid ejection head according to Japanese Patent Laid-Open No. 2003-312006 includes a separator structure to separate air and a liquid from each other, and an air vent region. Accordingly, this liquid ejection head creates a concern such as an increase in size of the head and ink adherence to the separator structure. Meanwhile, bubbles are guided to the air-liquid separator structure by inclining inside of a circulation path. However, this circulation path does not pass through the inside of a pressure chamber of the print head including nozzles to eject the liquid. In

other words, according to the technique of Japanese Patent Laid-Open No. 2003-312006, there is no circulation of the fluid in the pressure chamber. As a consequence, there is a risk of causing an ejection error in a case where bubbles and the like enter the pressure chamber, for example.

SUMMARY

(4) Applicant's disclosure provides a liquid ejection head and a liquid ejection apparatus, which suppress the occurrence of an ejection error without increasing a size of the apparatus.

(5) According to an aspect of the present disclosure, a liquid ejection apparatus includes a printing element board including a pressure chamber provided with an ejection port, wherein the printing element board is configured to eject a liquid from the ejection port, a supply flow passage provided to the printing element board and communicating with the pressure chamber, a collection flow passage provided to the printing element board and communicating with the pressure chamber, a liquid feeding mechanism configured to generate a difference in pressure between the supply flow passage and the collection flow passage in such a way as to supply the liquid from the supply flow passage to the pressure chamber and to recover the liquid in the pressure chamber from the collection flow passage, a first bubble reservoir unit connecting the supply flow passage to the liquid feeding mechanism, and a second bubble reservoir unit connecting the collection flow passage to the liquid feeding mechanism, wherein a volume of the first bubble reservoir unit is larger than a volume of the second bubble reservoir unit.

(6) According to the present disclosure, it is possible to provide a liquid ejection head and a liquid ejection apparatus, which suppress the occurrence of an ejection error without increasing a size of the apparatus.

(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic perspective view of a liquid ejection apparatus that can apply a liquid ejection head;

(2) FIG. **2** is a perspective view of the liquid ejection head;

(3) FIG. **3** is an exploded perspective view of the liquid ejection head;

(4) FIG. **4** is a schematic diagram showing a circulation path for one ink color in a steady state;

(5) FIG. **5**A is a cross-sectional view of a printing element board taken at a certain position in y direction;

(6) FIG. **5**B is a cross-sectional view of the printing element board taken at a different position in the y direction;

(7) FIG. **5**C is a cross-sectional view of the printing element board taken at another different position in the y direction;

(8) FIG. **6** is a diagram showing a flow of an ink in a case of performing printing by using a majority of ejection ports;

(9) FIG. **7** is a side view showing the liquid ejection head;

(10) FIG. **8**A is a cross-sectional view showing the liquid ejection head;

(11) FIG. **8**B is another cross-sectional view showing the liquid ejection head;

(12) FIG. **9** is a schematic diagram that clarifies inside of a circulation unit;

(13) FIG. **10**A is a cross-sectional view showing a first ink connection flow passage and a second ink connection flow passage;

(14) FIG. **10**B is another cross-sectional view showing the first ink connection flow passage and the second ink connection flow passage;

(15) FIG. **11**A is a schematic diagram showing a circulation flow passage;

(16) FIG. **11**B is another schematic diagram showing the circulation flow passage;
(17) FIG. **12**A is a diagram showing a flow of the ink and behaviors of bubbles in a case of using numerous ejection ports;
(18) FIG. **12**B is another diagram showing a flow of the ink and behaviors of bubbles in the case of using numerous ejection ports;
(19) FIG. **13**A is a cross-sectional view showing a first bubble reservoir unit and a second bubble reservoir unit;
(20) FIG. **13**B is another cross-sectional view showing the first bubble reservoir unit and the second bubble reservoir unit;
(21) FIG. **14** is a cross-sectional view taken along the XIV-XIV line in FIG. **13**A;
(22) FIG. **15** is a diagram showing a cross-section taken along the XV-XV line in FIG. **7**;
(23) FIG. **16**A is a schematic diagram showing the circulation flow passage;
(24) FIG. **16**B is another schematic diagram showing the circulation flow passage;
(25) FIG. **17**A is a diagram showing an example of a pressure adjustment unit;
(26) FIG. **17**B is another diagram showing the example of the pressure adjustment unit;
(27) FIG. **17**C is another diagram showing the example of the pressure adjustment unit;
(28) FIG. **18**A is an external perspective view of a circulation pump;
(29) FIG. **18**B is another external perspective view of the circulation pump;
(30) FIG. **19** is a cross-sectional view of the circulation pump taken along the XIX-XIX line;
(31) FIG. **20**A is a diagram for explaining a flow of an ink in the liquid ejection head;
(32) FIG. **20**B is another diagram for explaining the flow of the ink in the liquid ejection head;
(33) FIG. **20**C is another diagram for explaining the flow of the ink in the liquid ejection head;
(34) FIG. **20**D is another diagram for explaining the flow of the ink in the liquid ejection head;
(35) FIG. **20**E is another diagram for explaining the flow of the ink in the liquid ejection head;
(36) FIG. **21**A is a schematic diagram showing a circulation path for one ink color in an ejection unit;
(37) FIG. **21**B is another schematic diagram showing the circulation path for one ink color in the ejection unit;
(38) FIG. **22** is a diagram showing an opening plate;
(39) FIG. **23** is a diagram showing an ejection element board;
(40) FIG. **24**A is a cross-sectional view showing a flow of an ink at a certain portion of the ejection unit;
(41) FIG. **24**B is a cross-sectional view showing the flow of the ink at a different portion of the ejection unit;
(42) FIG. **24**C is a cross-sectional view showing the flow of the ink at another different portion of the ejection unit;
(43) FIG. **25**A is a cross-sectional view showing a portion in the vicinity of an ejection port in an ejection module;
(44) FIG. **25**B is another cross-sectional view showing the portion in the vicinity of the ejection port in the ejection module;
(45) FIG. **26** is a diagram showing an ejection element board of a comparative example;
(46) FIG. **27**A is a diagram showing a flow passage configuration of a liquid ejection head adaptable to inks of three colors;
(47) FIG. **27**B is another diagram showing the flow passage configuration of the liquid ejection head adaptable to inks of three colors; and
(48) FIG. **28** is a diagram showing a state of connection of the liquid ejection head to an ink tank and an external pump.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

(49) A first embodiment of the present disclosure will be described below with reference to the

drawings.

(50) FIG. **1** is a schematic perspective view of a liquid ejection apparatus **2000** that can apply a liquid ejection head **1000** according to the present embodiment. The liquid ejection apparatus **2000** of the present embodiment is an ink jet printing apparatus of a serial scanning type, which is configured to print an image on a print medium P by ejecting liquids (hereinafter also referred to as inks) from liquid ejection heads **1000** and **1001**. The liquid ejection heads **1000** and **1001** can be mounted on a carriage **10**, and the carriage **10** is provided to be movable in a main scanning direction being x direction along a guide shaft **11**. The print medium P is transported in a vertical scanning direction being y direction intersecting with (at right angle in the present embodiment) the main scanning direction by using a not-illustrated transportation roller.

(51) Two types of liquid ejection heads are mounted on the carriage **10**. The liquid ejection head **1000** can eject three types of inks while the liquid ejection head **1001** can eject six types of inks. The inks are pressure-supplied from nine types of ink tanks (liquid tanks) **2** (**21**, **22**, **23**, **24**, **25**, **26**, **27**, **28**, and **29**) to the liquid ejection heads through ink supply tubes **30**, respectively. A supply pump to be described later for the pressure supply is mounted on an ink supply unit **12**.

(52) As a modified example, the ink tanks can be reduced to seven types by setting the three types of the inks in the liquid ejection head **1000** to the ink of the same type, or the liquid election apparatus can be configured to be able to eject twelve or more types of inks by additionally mounting one or more liquid ejection heads.

(53) The liquid ejection head **1000** is fixed to and supported by the carriage **10** by using a positioning unit and electric contact points of the carriage **10**. The liquid ejection head **1000** performs printing by ejecting the inks while being moved in the scanning direction being the x direction.

(54) FIG. **2** is a perspective view of the liquid ejection head **1000** of the present embodiment, and FIG. **3** is an exploded perspective view of the liquid ejection head **1000**. The liquid ejection head **1000** includes a printing element unit **100**, a circulation unit **200**, a head housing unit **300**, and a cover **502**. The printing element unit **100** includes a printing element board **110**, a support member (a flow passage member) **102** provided with an ink supply connection flow passage **310** and an ink collection connection flow passage **320** connected to the printing element board **110**, an electric wiring tape **103**, and an electric contact board **104**. The electric contact board **104** includes electric contact points with the carriage **10**, and supplies signals and energy for driving a circulation pump **203** mounted on the circulation unit **200** through a circulation unit connector **106** and not-illustrated pump lines. Moreover, the electric contact board **104** supplies driving signals and energy for ink ejection to the printing element board **110** through the electric wiring tape **103**.

(55) An electric connection module is embodied by using an anisotropic conductive film (not shown), wire bonding, solder mounting and the like. However, the connecting method is not limited only to these methods. In the present embodiment, the connection between the printing element board **110** and the electric wiring tape **103** is carried out by wire bonding. The electric connection module is sealed with a sealing material so as to protect the electric connection module against corrosion with the inks and against external impacts.

(56) The circulation unit **200** includes a first pressure adjustment mechanism **201** and a second pressure adjustment mechanism **202** (see FIG. **4** to be described later) which are capable of adjusting pressure in a circulation path, and the circulation pump **203**. The inks are supplied from the ink tanks **2** to ink supply ports **32** through the ink supply tubes **30** (see FIG. **1**) and the head housing unit **300** provided with tube connectors **31**. In the present embodiment, ink supply passages are formed by fixing the circulation unit **200** to the head housing unit **300** by using screws **501**.

(57) An elastic member such as rubber and elastomer is employed as a sealing member used in a connector in each ink supply passage. The printing element unit **100** is attached and fixed to the head housing unit **300**, thus constituting the ink supply passage. The elastic member may be used in

the connector in the ink supply passage. The head housing unit **300** is formed from a combination of components obtained by injection molding of a resin containing filler in order to achieve positioning relative to the carriage **10** and to form a shape of an ink flow passage.

(58) The printing element board **110** is provided with ejection port rows by arranging ejection ports in the y direction. The multiple ejection port rows are provided in the x direction.

(59) FIG. **4** is a schematic diagram showing a circulation path for one ink color in a steady state to be applied to the liquid ejection apparatus **2000** of the present embodiment. The ink is pressure-supplied from the ink tank **21** to the liquid ejection head **1000** by using a supply pump P**0**. The ink is deprived of dust and the like by using a filter **204**, and is then supplied to the first pressure adjustment mechanism **201**. In FIG. **4** (as well as in FIG. **6** to be described later), the first pressure adjustment mechanism **201** is marked with "L" while the second pressure adjustment mechanism **202** is marked with "H". Here, "H" represents a high negative pressure and "L" represents a low negative pressure, which are opposite from high and low levels based on a positive pressure. The first pressure adjustment mechanism **201** adjusts a pressure in a first pressure control chamber **211** to a predetermined pressure (a negative pressure). The circulation pump **203** is a piezoelectric diaphragm pump configured to change a volume inside a pump chamber by inputting a driving voltage to a piezoelectric element attached to a diaphragm, and to feed a liquid by alternately activating two check valves along with pressure variations.

(60) The circulation pump **203** feeds the ink from a second pressure control chamber **221** on a low pressure (a high negative pressure) side to the first pressure control chamber **211** on a high pressure (a low negative pressure) side. The second pressure control chamber **221** is subjected to pressure adjustment to the lower pressure than that in the first pressure control chamber **211** by the second pressure adjustment mechanism **202**. Pressure chambers **113** each having an ejection port that can eject the liquid are disposed on the printing element board **110**. Common supply flow passages **111** and common collection flow passages **112** are connected to the respective pressure chambers **113**.

(61) Each common supply flow passage **111** is connected to the first pressure control chamber **211** through the first ink connection flow passage **310**, and the pressure in the common supply flow passage **111** is therefore adjusted to a high pressure (an upstream) side. Each common collection flow passage **112** is connected to the second pressure control chamber **221** through the second ink connection flow passage **320**, and the pressure in the common collection flow passage **112** is therefore adjusted to a low pressure (a downstream) side. A flow in a direction of an arrow α in FIG. **4** is generated in each pressure chamber **113** due to a difference in pressure between the common supply flow passage **111** and the common collection flow passage **112**. A portion of the ink with increased viscosity, which is present in the vicinity of each ejection port that is in a standby state or not ejecting the ink during a printing operation, is recovered from the pressure chamber **113**. Thus, an ejection error can be suppressed.

(62) In the present embodiment, a first bubble reservoir unit **301** is disposed in the first ink connection flow passage **310**, and a second bubble reservoir unit **302** is disposed in the second ink connection flow passage **320**. Each of the first bubble reservoir unit **301** and the second ink connection flow passage **320** has such a volume that can temporarily reserve bubbles inside ink paths which are generated either during the printing operation or during the standby for printing.

(63) FIGS. **5**A to **5**C are cross-sectional views of the printing element board **110** taken at different positions in the y direction. The printing element board **110** includes a Si substrate **120** on which a not-illustrated electric circuit and heaters **115** serving as pressure generating mechanisms are disposed, and an ejection port member **130** in which the pressure chambers **113** and ejection ports **114** corresponding to the heaters **115** are patterned by photolithography. Although the present embodiment is configured to obtain ejection energy by generating a bubble of the ink inside each pressure chamber **113** by applying a voltage to the corresponding heater **115**, the pressure generating mechanism is not limited only to this configuration. A piezoelectric element may be used instead of the heater. The printing element board **110** is stacked on the support member **102**

while the Si substrate **120** includes a contact surface **123**. The contact surface **123** is attached and fixed to the support member **102**, thereby being connected to the respective ink supply passages.
(64) In the present embodiment, distances in the x direction of the common supply flow passages **111** and the common collection flow passages **112** are set to a pitch of 1 mm or below in order to ensure ink supply performances to the pressure chambers **113** and to achieve cost reduction by downsizing the substrate size. Meanwhile, four ejection port rows each arranging the ejection ports at 600 dpi are deployed from the viewpoint of dotting efficiency on the print medium P. Note that the resolution of the ejection port deployment and the number of ejection port rows are not limited to the aforementioned examples.
(65) FIG. **5**A shows a cross-section of common supply flow passage openings **121** at a position where the common supply flow passages **111** communicate with the contact surface **123**. FIG. **5**B shows a cross-section at a position where none of the common supply flow passages **111** and the common collection flow passages **112** communicate with the contact surface **123**. FIG. **5**C shows a cross-section of common collection flow passage opening **122** at a position where the common collection flow passages **112** communicate with the contact surface **123**.
(66) In order to control a difference in pressure between each common supply flow passage **111** and the corresponding common collection flow passage **112**, it is necessary to divide the ink supply passages other than the pressure chambers **113** and the pressure control mechanisms. To this end, the first ink connection flow passage **310** and the second ink connection flow passage **320** need to be divided in a direction of the ejection port rows at the position of the cross-section shown in FIG. **5**B. Each of the common supply flow passage **111** and the common collection flow passage **112** has a very small cross-sectional area, and therefore has a risk of causing a shortage of ink supply due to a pressure loss associated with liquid feeding. For this reason, it is desirable to form the common supply flow passage **111** and the common collection flow passage **112** not communicating with the contact surface **123** shown in FIG. **5**B as short as possible. Accordingly, it is desirable to provide the numerous common supply flow passage openings **121** shown in FIG. **5**A and numerous common collection flow passage openings **122** shown in FIG. **5**C in the direction of the ejection port rows.
(67) In the exploded perspective view of FIG. **3**, the first ink connection flow passages **310** for one color are disposed at nine positions and the second ink connection flow passages **320** for the one color are disposed at eight positions. The numbers of these positions of connection vary depending on lengths of the ejection port rows and on a division bonding width. In the present embodiment, the cross-sectional area of each common supply flow passage **111** and each common collection flow passage **112** of FIG. **5**B is equal to or below 0.1 mm.sup.2, and a distance between each common supply flow passage opening **121** and the corresponding common collection flow passage opening **122** is equal to or below 7.5 mm.
(68) FIG. **6** shows a flow of the ink in the circulation path for one color in a case of performing printing by using the majority of the ejection ports. In the case of performing the printing by using the majority of the ejection ports, the ink is supplied from both of the common supply flow passage **111** and the common collection flow passage **112** to the corresponding pressure chamber **113** unlike the way of flow in the case of circulation in the steady state.
(69) In the case where the ink is ejected from a certain pressure chamber **113**, the ink is supplied from each of the common supply flow passage **111** and the common collection flow passage **112**. The common supply flow passage **111** supplies the ink, which is supplied from the first pressure control chamber **211** through the first ink connection flow passage **310**, to the corresponding pressure chamber **113**. Meanwhile, the common collection flow passage **112** supplies the ink, which is supplied from the second pressure control chamber **221** through the second ink connection flow passage **320**, to the corresponding pressure chamber **113**. The circulation pump **203** transports the ink from the second pressure control chamber **221** to the first pressure control chamber **211** in the same way as in the steady state.

(70) In this instance, the second pressure control chamber **221** supplies the ink to the second ink connection flow passage **320** and the circulation pump **203**. Moreover, the second pressure control chamber **221** retains the constant pressure by causing the second pressure adjustment mechanism **202** to supply the ink from the first pressure control chamber **211** through a bypass flow passage that connects the first pressure adjustment mechanism **201** to the second pressure adjustment mechanism **202**. While the first pressure control chamber **211** supplies the ink to the second pressure adjustment mechanism **202** and the first ink connection flow passage **310**, the first pressure control chamber **211** retains the constant pressure by causing the first pressure adjustment mechanism **201** to recover the ink from the ink tank **21** serving as an ink supply source together with the portion of the ink transported by the circulation pump **203**.
(71) As described above, the direction of flow of the ink in the common collection flow passage **112** is changed depending on the printing state, and the direction of flow of the ink in the second ink connection flow passage **320** is changed in accordance therewith.
(72) FIG. **7** is a side view showing the liquid ejection head **1000**. FIG. **8**A is a cross-sectional view taken along the VIIIA-VIIIA line in FIG. **7**. FIG. **8**B is a cross-sectional view taken along the line in FIG. **7**. The printing element board **110** is provided with the ejection port rows along the y direction being a direction of movement of the print medium P, and the inks are ejected in z direction from the respective ejection ports. The first ink connection flow passage **310** and the second ink connection flow passage **320** are formed from the head housing unit **300** and the support member **102**.
(73) The printing element board **110** is supported by the support member **102**. The printing element board **110** is supported in such a way as to establish connection from the first pressure control chamber **211** to the common supply flow passage opening **121** and the common supply flow passage **111** through the first ink connection flow passage **310**. Meanwhile, the printing element board **110** is supported in such a way as to establish connection from the second pressure control chamber **221** to the common collection flow passage opening **122** and the common collection flow passage **112** through the second ink connection flow passage **320** as shown in FIG. **8**B.
(74) The first pressure control chamber **211** and the second pressure control chamber **221** are controlled at constant pressures by using the pressure adjustment mechanisms built in the circulation unit **200**.
(75) FIG. **9** is a schematic diagram that clarifies the inside of the circulation unit **200**. In the circulation unit **200**, the ink is pressure-supplied from the ink supply unit **12** to the first pressure adjustment mechanism **201** through the ink supply port **32** and the filter **204**. The pressure adjustment mechanism **201** includes a valve **232**, a valve spring **233**, a flexible member **231**, a pressing plate **235**, and a pressure adjustment spring **234**.
(76) In the pressure control chamber **211**, the pressing plate **235** deforms the flexible member **231** and the pressure adjustment spring **234** in the case where a volume of the pressure control chamber **211** is reduced due to discharge of the ink and the like, thus attempting to keep the pressure inside the pressure control chamber **211** constant. By compressive deformation of the pressure adjustment spring **234**, the valve spring **233** is deformed in a compressive direction through the valve **232**. Thus, it is possible to open the valve **232** and to supply the ink to the pressure control chamber **211**. This behavior makes it possible to supply the ink and to keep the constant pressure inside the pressure control chamber **211**. The negative pressure in the pressure control chamber **211** is set depending on positions of contact of the pressure adjustment spring **234** and the valve **232** with the pressing plate **235**.
(77) The pressure adjustment mechanism **202** of the pressure control chamber **221** includes a valve **242**, a valve spring **243**, a flexible member **241**, a pressing plate **245**, and a pressure adjustment spring **244**. The principle of adjustment of the pressure in the pressure adjustment mechanism **202** is the same as the principle applicable to the pressure adjustment mechanism **201** with the only exception that the ink supply source is changed from the ink supply unit **12** to the pressure control

chamber **211**.

(78) The circulation pump **203** is connected in such a way as to feed the ink in the pressure control chamber **221** to the pressure control chamber **211**. In the present embodiment, a small diaphragm pump using a piezoelectric element is adopted as the circulation pump **203**. Since the pump can be driven by applying a voltage pulse to the piezoelectric element, it is possible to control on and off of the circulation pump **203** by using the inputted voltage pulse. By transferring the ink in the pressure control chamber **221** to the pressure control chamber **211** by using the circulation pump **203**, the pressure control chamber **211** is set to a state of an applied pressure in an amount equivalent to the ink that is fed in, and the pressure control chamber **221** is set to a state of a negative pressure in an amount equivalent to the ink that is fed out.

(79) As the pressure control chamber **221** is set to the negative pressure, the pressure control chamber **221** recovers the ink through the pressure adjustment mechanism **202**. On the other hand, the pressure adjustment mechanism **202** recovers the ink from the pressure control chamber **211** and the pressure chamber **113**, and therefore generates a circulating flow while keeping the pressure constant. As a consequence of generating the circulating flow through the pressure chamber **113** as described above, it is possible to remove the ink with increased viscosity in the vicinity of the ejection ports due to evaporation of the ink, and thus to achieve stable ejection.

(80) FIG. **10**A is a cross-sectional view showing the first ink connection flow passage **310** connected to the pressure control chamber **211** in the present embodiment, and FIG. **10**B is a cross-sectional view showing the second ink connection flow passage **320** connected to the pressure control chamber **221**. The printing element board **110** includes the ejection port member **130** and the Si substrate **120**. A not-illustrated warming heater for stabilizing ejection is disposed on the Si substrate **120**. Meanwhile, in order to homogenize the temperature of the entire printing element board **110** and to achieve stable bonding to the Si substrate **120**, the support member **102** employs an alumina material that has high thermal conductivity and a linear coefficient of expansion which is close to that of Si.

(81) In FIGS. **10**A and **10**B, arrows (in solid lines) indicated in the flow passages show flows of the circulated inks by driving the circulation pump **203** at the time of not performing the printing. To be more precise, in FIG. **10**A, the ink flows from the pressure control chamber **211** to the common supply flow passage opening **121** while passing through the first ink connection flow passage **310** which is formed from the head housing unit **300** including the first bubble reservoir unit **301** and from the support member **102**. This flow of the ink starts from the common supply flow passage **111**, passes through the pressure chambers **113** that eject the ink, flows to the common collection flow passage **112**, and is then recovered from the common collection flow passage opening **122**. Moreover, the second ink connection flow passage **320** which is formed from the head housing unit **300** including the second bubble reservoir unit **302** and from the support member **102** supplies the ink recovered from the common collection flow passage opening **122** to the pressure control chamber **221**. Subsequently, the circulation pump **203** feeds the ink from the pressure control chamber **221** to the pressure control chamber **211**. Thus, the circulating flow goes around.

(82) FIG. **11**A is a schematic diagram showing the circulation flow passage in a state of ink ejection, and FIG. **11**B is a schematic diagram showing the circulation flow passage in a case of continuing the circulation for a while without generation of bubbles. The circulating flow is completed inside the ink flow passages of the liquid ejection head **1000**. Accordingly, bubbles **500** generated inside the flow passages of the liquid ejection head **1000** should be present somewhere in the circulating flows inclusive of the circulation in the first pressure control chamber **211** and in the second pressure control chamber **221**. The bubbles **500** are generated by: foaming caused at the time of ink filling, along with the ink flow, and the like; supersaturation of a gas dissolved in the ink associated with a rise in temperature or reduction in pressure inside the liquid ejection head **1000**; and the like. In the case where the bubbles **500** flow into any of the pressure chambers **113**, the bubbles **500** are prone to cause an ejection error of the ink that may lead to an image error.

Accordingly, it is desirable to reserve these bubbles **500** at a portion of the circulation flow passage located away from the pressure chambers **113** so as not to let the bubbles **500** flow into the pressure chambers **113**.
(83) In a case of a general liquid ejection head without provision of a portion to reserve bubbles in its flow passage, it is necessary to use the liquid ejection head in a range where the dissolved gas does not cause supersaturation while controlling a degree of deaeration of the ink, or to discharge generated bubbles out of the liquid ejection head in each case of generation. There are methods of controlling the degree of deaeration including agitation under a reduced pressure, a deaeration module adopting a hollow fiber membrane, and the like. However, these methods may cause high costs and increases in head size and weight, and are therefore likely to adversely affect a printing speed and other performances. On the other hand, in the case where the ink containing the bubbles is discharged in each case, the ink supposed to be used for printing is discharged as waste ink. Therefore, this method leads to an increase in printing cost.
(84) Given the circumstances, in the present embodiment, as shown in FIG. **11**A, the first bubble reservoir unit **301** and the second bubble reservoir unit **302** are located at positions distant from the pressure chambers **113** in the circulation flow passage so as to gather the generated bubbles **500** in the first bubble reservoir unit **301** and the second bubble reservoir unit **302**. In this way, it is possible to keep the bubbles **500** from flowing into the pressure chambers **113**, thereby reducing the chance of causing an ejection error. By reducing the occurrence of the ejection error in accordance with the above-described method, it is possible to suppress a significant increase in head size or an increase in amount of waste ink.
(85) Due to the circulating flow at the time of not performing the printing, the flow of the ink from the printing element board **110** inclusive of the pressure chambers **113** toward the second bubble reservoir unit **302** is generated in the second ink connection flow passage **320**. Accordingly, the bubbles **500** are gathered in the second bubble reservoir unit **302** by the flow of the ink. On the other hand, the flow of the ink directed to the printing element board **110** is generated in the first ink connection flow passage **310**. It is therefore difficult to gather the bubbles **500** in the first bubble reservoir unit **301**.
(86) Given the circumstances, ceiling surfaces in the first ink connection flow passage **310** have angles (θ**11** and θ**13**) in a range from about 40 degrees to 50 degrees relative to the surface provided with the ejection ports in the present embodiment (see FIG. **10**A). Here, the ceiling is a surface that defines part of the flow passage, which corresponds to an inner wall of the flow passage where a component force of a normal vector at the ceiling surface has a component in a gravitational direction (the z direction).
(87) According to the above-described configuration, even in the case where the flow of the ink directed to the printing element board **110** is generated, the bubbles **500** are guided easily to the first bubble reservoir unit **301** so that the bubbles **500** can be gathered at the position distant from the pressure chambers **113**. These angles θ are determined based on a coefficient of friction defined by physical properties of the ink and the inner wall of the first ink connection flow passage **310**, and on a migration force attributed to buoyancy.
(88) It has been confirmed that the ink used in the liquid ejection head **1000** and the member of the first ink connection flow passage **310** in the present embodiment successfully achieved the effects of the present embodiment by providing the ceiling surfaces with the angle of about 15 degrees or above relative to the surface provided with the ejection ports. It is more preferable to set each ceiling surface with an angle close to 90 degrees with which 100% of the component force of the buoyancy of each bubble **500** can be used for the migration force.
(89) Moreover, in the present embodiment, ceiling surfaces in the second ink connection flow passage **320** have angles (θ**22** and θ**24**) in a range from about 40 degrees to 50 degrees relative to the surface provided with the ejection ports in the present embodiment (see FIG. **10**B). Accordingly, the movement of the bubbles **500** to the second bubble reservoir unit **302** can be

completed in a short time by using a circulating flow pressure in addition to the migration force attributed to the buoyancy.

(90) Meanwhile, the bubbles **500** gathered in the second bubble reservoir unit **302** gradually move from the second bubble reservoir unit **302** to the first pressure adjustment chamber **211** and the first bubble reservoir unit **301** through the second pressure adjustment chamber **221** and the circulation pump **203** by use of the circulating flow (see FIG. **11**B). The bubbles **500** in the first bubble reservoir unit **301** also have to be kept from reaching the pressure chambers **113** in the case where the bubbles **500** move toward the supply flow passage on the downstream of the circulation pump as mentioned above. For this reason, it is desirable to secure a volume large enough for reserving the bubbles **500** by setting the volume on the downstream of the circulation pump **203** larger than that of the flow passage on the upstream of the circulation pump **203**.

(91) Meanwhile, in the circulation passage, the first pressure control chamber **211** and the second pressure control chamber **221** can change their volumes for the purpose of the pressure control. It is therefore desirable to provide the bubble reservoir units of certain volumes at separate locations. In the present embodiment, the first bubble reservoir unit **301** and the second bubble reservoir unit **302** are provided, and the volumes of these units are set to satisfy a relation of (volume of first bubble reservoir unit **301**)>(volume of second bubble reservoir unit **302**).

(92) Here, a capability of reserving the bubbles **500** is higher as a flow passage volume is larger. However, a careless increase of the entire volume may lead to an increase in head size or an increase in amount of the required ink. Meanwhile, in a case of a complicated flow passage shape provided with a space in a certain volume, there is a filling method called choke suction, which is designed to perform ink filling by releasing an ink supply valve after reducing a pressure to a certain pressure so as to fill a space in an anti-gravitational direction with the ink as well.

(93) While the present embodiment also employs the choke suction, this filling method still causes the gas to be left at a certain percentage of the entire flow passage volume. Accordingly, an expansion in flow passage volume also leads to an increase in initial amount of the gas. For this reason, the larger flow passage volume is not always desirable, and the total flow passage volume and the difference in volume between the upstream and the downstream of the circulation pump need to be determined in consideration of the pressure loss, the initial amount of the gas, a generated amount of the gas, and so forth.

(94) In the case of the present embodiment, the volume on the downstream of the pump (from an outlet of the pump to the ejection ports) is equal to about 6.4 cc while the volume on the upstream of the pump (from the ejection ports to an inlet of the pump) is equal to about 4.2 cc. Here, the volume on the downstream of the pump is nearly 1.5 times as large as the volume on the upstream of the pump. These volumes create a difference in volume while summing up the first and second pressure adjustment chambers, the first and second ink connection flow passages, and the flow passages connecting these constituents to one another. In other words, a relation of “volume of first pressure adjustment chamber (**211**)>volume of second pressure adjustment chamber (**221**)” holds true and a relation of “volume of first ink connection flow passage (**310**)>volume of second ink connection flow passage (**320**)” holds true.

(95) The ink in an amount of about 20% of the entire volume is calculated to remain even in the case where the entire volume on the upstream of the pump is gasified and moves to the downstream of the pump due to the remaining gas of initial ink filling and generation of the gas associated with continuous ejection. An initial ink filling ratio relies on product specifications depending on the pressure loss at the time of filling and the number of times of suctioning. The amount of generation of the gas due to the ejection varies depending on the ink type and the temperature. Moreover, the timing of discharge of the gas also relies on the product specifications. Accordingly, it is not possible to generally determine a minimum volume rate. However, it is necessary to set the volume on the downstream of the pump at least about 1.2 times as large as the volume on the upstream of the pump in order to cause at least 10% of the ink to remain.

(96) In the present embodiment, the pump is mounted inside the liquid ejection head **1000**. This is a particularly effective configuration because a method of removing the gas in the flow passages is limited to suctioning from the ejection ports. However, the present embodiment is not limited only to the configuration to complete the circulation flow passages inside the liquid ejection head. For example, a case of providing the circulation pump outside the liquid ejection head (in a printing apparatus body or the like) and a configuration not provided with a gas removal unit are also acceptable. Even in the case of the above-mentioned configuration, it is effective to set a "volume from a discharge port of a circulation pump to supply ports of a head and a printing element board" larger than a "volume from collection ports of the head and the printing element board to an inflow port of the circulation pump" in order to reduce a frequency of gas removal operations.
(97) Meanwhile, as a consequence of setting the volume of the first ink connection flow passage **310** larger than the volume of the second ink connection flow passage **320**, a flow velocity of the ink flowing in the first ink connection flow passage **310** becomes slower than a flow velocity of the ink flowing in the second ink connection flow passage **320**. By slowing down the flow velocity of the ink, the bubbles **500** in the first ink connection flow passage **310** are easily detached from the pressure chambers **113** thanks to the action of buoyancy. In addition, by setting the volume of the second ink connection flow passage **320** smaller than the volume of the first ink connection flow passage **310**, the flow velocity of the ink flowing in the second ink connection flow passage **320** becomes faster than the flow velocity of the ink flowing in the first ink connection flow passage **310**. By increasing the flow velocity of the ink, the bubbles **500** in the second ink connection flow passage **320** are detached easily from the pressure chambers **113** due to the action of buoyancy and the flow velocity of the ink.
(98) FIGS. **12**A and **12**B are diagrams showing the flow of the ink and behaviors of the bubbles **500** in a case of using the majority of the ejection ports shown in FIG. **6**. FIG. **12**A is a cross-sectional view showing the first ink connection flow passage **310** connected to the pressure control chamber **211**, and FIG. **12**B is a cross-sectional view showing the second ink connection flow passage **320** connected to the pressure control chamber **221**. Positions of the cross-sections in FIGS. **12**A and **12**B are the same as those in FIGS. **10**A and **10**B.
(99) In the case of performing the printing by using the majority of the ejection ports, the ink in an amount larger than that in the circulating flow in the state of not performing the printing shown in FIGS. **10**A and **10**B is supplied to the pressure chambers **113**, whereby a large current is generated in each flow passage. Meanwhile, in the case of performing the printing by using the majority of the ejection ports, the circulating flow of the ink in each of the first ink connection flow passage **310** and the second ink connection flow passage **320** is the flow directed to the pressure chambers **113**. As a consequence of the increase in ink flow volume, the ink flow velocity is increased overall in the direction toward the pressure chambers **113**.
(100) Particularly, in the first ink connection flow passage **310** and the second ink connection flow passage **320** formed from the support member **102** having the relatively small cross-sectional area of each flow passage therein, a fast flow velocity is generated and a dynamic pressure to be applied to the bubbles **500** is increased, whereby the bubbles **500** are more likely to flow into the pressure chambers **113**. Meanwhile, in the case of the present embodiment, ejection energy in each pressure chamber **113** is generated by using thermal energy from the heater **115**. Accordingly, the temperature of the printing element board **110** is increased along with the ejection. As a consequence, the inside of the circulating flow passages formed in the support member **102** and the printing element board **110** reaches a relatively high temperature, and the gas dissolved in the ink is more likely to be supersaturated and prone to generate the bubbles **500**.
(101) In the case of performing the printing by using the majority of the ejection ports as described above, the bubbles **500** have to be moved to the first bubble reservoir unit **301** or the second bubble reservoir unit **302** by regularly establishing the state of circulation at the time of not performing the printing or by stopping the circulation depending on the amount of ejection or an ejection period of

the ink. This period to move the bubbles **500** may involve suspension of the printing as mentioned above, and may therefore deteriorate printing productivity. Accordingly, it is desirable to set each ceiling surface with an angle close to 90 degrees with which 100% of the component force of the buoyancy of the bubbles **500** can be used for the migration force in order to reduce the period to move the bubbles **500**.

(102) As a modified example, there is a case of mounting a heater on the printing element board **110** in order to adjust the temperature of the ink, and a resin material having low thermal conductivity may be used for the support member **102** while focusing on a temperature adjustment rate. In that case, a location where the bubbles are generated by the heat is limited to a portion in the vicinity of the Si substrate **120**.

(103) Meanwhile, the common supply flow passage **111** provided in the printing element board **110** is formed in accordance with Si substrate processing techniques. For this reason, it is difficult to secure a sufficient angle relative to the surface provided with the ejection ports. Moreover, since the cross-sectional area of each flow passage is very small, it is difficult to guide the bubbles **500** to the first bubble reservoir unit **301** by using the buoyancy against the circulating flow. Accordingly, it is necessary to discharge the bubbles **500** generated inside the common supply flow passage **111** regularly out of pressure chambers **113** through the ejection ports by suctioning and the like depending on the amount of ejection of the ink and on a printing period. Nonetheless, the volume of the ink in the common supply flow passage **111** is very small so that the amount of the waste ink can be minimized.

(104) FIG. **13**A is a cross-sectional view showing the first bubble reservoir unit **301** in the case of reserving a large amount of the bubbles **500**, and FIG. **13**B is a cross-sectional view showing the second bubble reservoir unit **302** in the case of reserving a large amount of the bubbles **500**. Positions of the cross-sections in FIGS. **13**A and **13**B are the same as those in FIGS. **10**A and **10**B. FIG. **14** is a cross-sectional view taken along the XIV-XIV line in FIG. **13**A, which is a view showing slits **303** in the first bubble reservoir unit **301** and the second bubble reservoir unit **302**. In the case where the bubbles **500** are bonded to one another into such a size that substantially occludes the cross-section of the flow passage, the bubbles **500** are drifted to the pressure chambers **113** due to an increase in drag attributed to the ink flow.

(105) However, the cross-sectional areas of the first bubble reservoir unit **301** and the second bubble reservoir unit **302** inclusive of the ceilings thereof are larger than a minimum cross-sectional area portion inside each bubble reservoir unit. In addition, each flow passage wall is provided with the slits **303** along the direction of flow of the ink. Each slit **303** is formed thin enough for not being clogged with the bubble **500**. As a consequence, a relative ink flow velocity in each bubble reservoir unit is slowed down so that the ink can be fed out of the slits **303** without moving the bubbles **500**. In this way, it is possible to keep the bubbles **500** from flowing into the pressure chambers **113**. In the present embodiment, each slit has a shape of a groove with a width in a range from 0.2 to 0.5 mm, and adopts such a structure that the bubbles **500** reserved and bonded together can hardly occlude the slits **303**.

(106) Even in the case where the slits **303** are provided as described above, a certain amount of the bubbles **500** may be reserved in the first bubble reservoir unit **301** and the second bubble reservoir unit **302**. In the case where the bubbles **500** reach the flow passage with the small cross-sectional area that increases the flow velocity, the bubbles **500** are prone to flow into the pressure chambers **113** due to the ink dynamic pressure, thereby causing an ejection error. For this reason, in the case where a certain amount of the bubbles **500** are reserved, it is necessary to conduct a recovery operation such as suctioning from the ejection ports in order to discharge the bubbles **500** to the outside. A suction recovery device and the like configured to perform a recovery operation by suctioning and so forth is a structure that has been widely adopted by ink jet printers for obtaining printing stability. This is not a new structure for removing the bubbles **500** stored in the first bubble reservoir unit **301** and the second bubble reservoir unit **302**.

(107) FIG. **15** is a diagram showing a cross-section taken along the XV-XV line in FIG. **7**. The first bubble reservoir unit **301** and the second bubble reservoir unit **302** can move the generated bubbles to the ceilings by securing the cross-sectional areas as large as possible. Accordingly, it is desirable to form the first bubble reservoir unit **301** and the second bubble reservoir unit **302** with the increased cross-areas of the flow passages to portions in the vicinity of the printing element board **110** where the bubbles **500** are prone to be generated.
(108) In the case of alternately disposing the common supply flow passage openings **121** at nine positions in the direction of the ejection port rows and the common collection flow passage openings **122** at eight positions in the same direction as in the present embodiment, the respective openings are joined to one another by using a flow passage that has a length equivalent to a long side in the y direction equal to or more than a length between two end portions of each ejection port row. In this case, branches need to be disposed for the purpose of supply to the respective openings arranged at narrow pitches. In the present embodiment, as shown in the cross-sectional views of FIGS. **8**A and **8**B, each portion to be connected to the printing element board **110** is formed into a branch unit having a triangular shape provided with an oblique side that is inclined in the x direction being the scanning direction. The oblique side in the triangular shape of the first ink connection flow passage **310** to be connected to the common supply flow passage opening **121** and the oblique side in the triangular shape of the second ink connection flow passage **320** to be connected to the common collection flow passage opening **122** are arranged in opposite directions to each other.
(109) As described above, there are provided the first bubble reservoir unit that connects a liquid feeding mechanism to the supply flow passage communicating with the pressure chamber, and the second bubble reservoir unit that connects the liquid feeding mechanism to the collection flow passage communicating with the pressure chamber, and the volume of the first bubble reservoir unit is set larger than the volume of the second bubble reservoir unit. Thus, it is possible to provide the liquid ejection head and the liquid ejection apparatus, which suppress the occurrence of an ejection error without increasing a size of the apparatus.
Second Embodiment
(110) A second embodiment of the present disclosure will be described below with reference to the drawings. Note that a basic configuration of the present embodiment is the same as the configuration of the first embodiment, and characteristic features of the present embodiment will therefore be discussed below.
(111) FIG. **16**A is a schematic diagram showing the circulation flow passage in the state of ink ejection, and FIG. **16**B is a schematic diagram showing the circulation flow passage in the case of continuing the circulation for a while without generation of bubbles. The liquid ejection head **1000** of the present embodiment includes a first bubble reservoir unit-cum-first pressure adjustment chamber **711** and a second bubble reservoir unit-cum-second pressure adjustment chamber **721**. In other words, each pressure adjustment chamber is also configured to serve as the bubble reservoir unit.
(112) The circulating flow is completed inside the ink flow passages of the liquid ejection head **1000**. Accordingly, the bubbles **500** generated inside the flow passages of the liquid ejection head **1000** should be present somewhere in the circulating flows. The bubbles **500** are generated by: foaming caused at the time of ink filling, along with the ink flow, and the like; supersaturation of a gas dissolved in the ink associated with a rise in temperature or reduction in pressure inside the liquid ejection head **1000**; and the like. In the case where the bubbles **500** flow into any of pressure chambers **613**, the bubbles **500** are prone to cause an ejection error of the ink that may lead to an image error. Accordingly, the bubbles **500** are temporarily reserved in the first bubble reservoir unit-cum-first pressure adjustment chamber **711** and the second bubble reservoir unit-cum-second pressure adjustment chamber **721** which are provided at positions located away from the pressure chambers **613** so as not to let the bubbles **500** flow into the pressure chambers **613**. Then, the

bubbles **500** are discharged out of the liquid ejection head **1000** by regular suctioning from the ejection ports.

(113) Meanwhile, the bubbles **500** reserved in the second bubble reservoir unit-cum-second pressure adjustment chamber **721** are moved by the circulating flow to the first bubble reservoir unit-cum-first pressure adjustment chamber **711** located downstream of the circulation pump (see FIG. **16**B). The volume on the downstream of the circulation pump is set larger than that of the flow passage on the upstream of the circulation pump so that the bubbles **500** in the first bubble reservoir unit-cum-first pressure adjustment chamber **711** can be kept from reaching the pressure chambers **613** even in the case where the bubbles **500** are moved to the supply flow passage side on the downstream of the circulation pump as mentioned above. It is desirable to increase the portion that can reserve the bubbles by increasing the volume as described above. Here, the larger flow passage volume increases the capability of reserving the bubbles **500**. Nevertheless, the larger flow passage volume is not always desirable, and the total flow passage volume and the difference in volume between the upstream and the downstream of the circulation pump need to be determined in consideration of the pressure loss, the initial amount of the gas, the generated amount of the gas, and so forth.

(114) Here, a connection flow passage between a common supply flow passage **611** and the first bubble reservoir unit-cum-first pressure adjustment chamber **711** as well as a communication flow passage between a common collection flow passage **612** and the second bubble reservoir unit-cum-second pressure adjustment chamber **721** are formed to extend in the vertical direction so that the bubbles **500** can move in the anti-gravitational direction by using buoyancy. Alternatively, each of the connection flow passages mentioned above preferably includes an inner wall in which a component force of a normal vector of a ceiling surface located vertically above has a component in the gravitational direction (the z direction).

REFERENCE EXAMPLES

(115) More detailed reference examples of the above-mentioned liquid ejection apparatus will be described.

Pressure Adjustment Unit

(116) FIGS. **17**A to **17**C are diagrams showing an example of a pressure adjustment unit. A configuration and operation of the pressure adjustment unit (a first pressure adjustment unit **1120** or a second pressure adjustment unit **1150**) that is built in the above-described liquid ejection head **1000** will be described in more detail with reference to FIGS. **17**A to **17**C. Note that the first pressure adjustment unit **1120** and the second pressure adjustment unit **1150** have substantially the same structure. Accordingly, the following description will be given of the first pressure adjustment unit **1120** as an example, and the second pressure adjustment unit **1150** will be explained by citing reference signs of portions corresponding to those of the first pressure adjustment unit in FIGS. **17**A to **17**C. In the case of the second pressure adjustment unit **1150**, a first valve chamber **1121** to be described below will be translated into a second valve chamber **1151**, and a first pressure control chamber **1122** to be described below will be translated into a second pressure control chamber **1152**.

(117) The first pressure adjustment unit **1120** includes the first valve chamber **1121** and the first pressure control chamber **1122** formed inside a cylindrical housing **1125**. The first valve chamber **1121** and the first pressure control chamber **1122** are separated from each other by a partition wall **1123** provided inside the cylindrical housing **1125**. Nevertheless, the first valve chamber **1121** communicates with the first pressure control chamber **1122** through a communication port **1191** formed in the partition wall **1123**. The first valve chamber **1121** is provided with a valve **1190** that switches between communication and block between the first valve chamber **1121** and the first pressure control chamber **1122** with the communication port **1191**. The valve **1190** is held at a position opposed to the communication port **1191** by using a valve spring **1200**, and the valve **1190** is configured to be capable of coming into close contact with the partition wall **1123** by using a

biasing force of the valve spring **1200**. As a consequence of bringing the valve **1190** into close contact with the partition wall **1123**, the flow of the ink through the communication port **1191** is blocked. Here, a portion of the valve **1190** to come into contact with the partition wall **1123** is preferably made of an elastic material in order to enhance close contact with the partition wall **1123**. Meanwhile, a valve shaft **1190***a* to be inserted into the communication port **1191** is provided in a projecting manner at a central portion of the valve **1190**. By pressing this valve shaft **1190***a* against the biasing force of the valve spring **1200**, the valve **1190** is detached from the partition wall **1123** so as to enable the ink to flow through the communication port **1191**. In the following description, a state where the flow of the ink through the communication port **1191** is blocked by the valve **1190** will be referred to as a "closed state" while a state where the ink can flow through the communication port **1191** will be referred to as an "open state".

(118) An opening of the cylindrical housing **1125** is occluded by a flexible member **1230** and a pressing plate **1210**. The flexible member **1230**, the pressing plate **1210**, peripheral walls of the housing **1125**, and the partition wall **1123** form the first pressure control chamber **1122**. The pressing plate **1210** is made displaceable along with displacement of the flexible member **1230**. Although materials of the pressing plate **1210** and the flexible member **1230** are not limited, it is possible to form the pressing plate **1210** from a resin molded component and to form the flexible member **1230** from a resin film, for example. In this case, the pressing plate **1210** can be fixed to the flexible member **1230** by heat sealing.

(119) A pressure adjustment spring **1220** (a biasing member) is provided between the pressing plate **1210** and the partition wall **1123**. As shown in FIG. **17**A, the pressing plate **1210** and the flexible member **1230** are biased in a direction to spread an inner volume of the first pressure control chamber **1122** by a biasing force of the pressure adjustment spring **1220**. Meanwhile, in a case where the pressure inside the first pressure control chamber **1122** is reduced, the pressing plate **1210** and the flexible member **1230** are displaced in a direction to reduce the inner volume of the first pressure control chamber **1122** against the pressure of the pressure adjustment spring **1220**. Then, in a case where the inner volume of the first pressure control chamber **1122** is reduced to a predetermined amount, the pressing plate **1210** comes into contact with the valve shaft **1190***a* of the valve **1190**. Thereafter, as the inner volume of the first pressure control chamber **1122** is reduced further, the valve **1190** moves together with the valve shaft **1190***a* against the biasing force of the valve spring **1200**, thereby being detached from the partition wall **1123**. In this way, the open state (a state in FIG. **17**B) of the communication port **1191** is established.

(120) In the present embodiment, connection setting in the circulation path is carried out such that the pressure in the first valve chamber **1121** is higher than the pressure in the first pressure control chamber **1122** in the case where the communication port **1191** is set to the open state. Hence, in the case where the communication port **1191** is set to the open state, the ink flows from the first valve chamber **1121** into the first pressure control chamber **1122**. This inflow of the ink displaces the flexible member **1230** and the pressing plate **1210** in the direction to increase the inner volume of the first pressure control chamber **1122**. As a consequence, the pressing plate **1210** is detached from the valve shaft **1190***a* of the valve **1190** and the valve **1190** comes into close contact with the partition wall **1123** by the biasing force of the valve spring **1200**. Thus, the closed state (a state in FIG. **17**C) of the communication port **1191** is established.

(121) As described above, according to the first pressure adjustment unit **1120** of the present embodiment, the ink flows in from the first valve chamber **1121** through the communication port **1191** in the case where the pressure inside the first pressure control chamber **1122** is reduced to the predetermined pressure or below (in the case where the negative pressure is increased, for example). Thus, the pressure in the first pressure control chamber **1122** is kept from being reduced further. In this way, the first pressure control chamber **1122** is controlled such that the pressure therein is maintained at the pressure within a prescribed range.

(122) Next, the pressure in the first pressure control chamber **1122** will be described further in

detail.

(123) Let us consider the state (the state in FIG. **17**B) where the flexible member **1230** and the pressing plate **1210** are displaced in accordance with the pressure in the first pressure control chamber **1122** as mentioned above, and the closed state of the communication port **1191** is established by the pressing plate **1210** coming into contact with the valve shaft **1190***a*. In this instance, a relation of forces that act on the pressing plate **1210** is expressed by the following formula 1:

$$P2\times S2+F2+(P1-P2)\times S1+F1=0 \qquad \text{Formula 1.}$$

(124) In addition, rearrangement of the formula 1 with respect to P2 gives the following formula:

$$P2=-(F1+F2+P1\times S1)/(S2-S1) \qquad \text{Formula 2,}$$

where P1: a pressure (a gauge pressure) in the first valve chamber **1121**, P2: a pressure (a gauge pressure) in the first pressure control chamber **1122**, F1: a spring force of the valve spring **1200**, F2: a spring force of the pressure adjustment spring **1220**, S1: a pressure receiving area of the valve **1190**, and S2: a pressure receiving area of the pressing plate **1210**.

(125) Here, regarding the spring force F1 of the valve spring **1200** and the spring force F2 of the pressure adjustment spring **1220**, a direction to press the valve **1190** and the pressing plate **1210** is determined to be positive (a leftward direction in FIG. **17**B). Meanwhile, regarding the pressure P1 in the first valve chamber **1121** and the pressure P2 in the first pressure control chamber **1122**, the pressure P1 is configured to satisfy a relation $P1\geq P2$.

(126) The pressure P2 in the first pressure control chamber **1122** in the case where the open state of the communication port **1191** is established is determined by the formula 2. In the case where the open stat of the communication port **1191** is established, the ink flows from the first valve chamber **1121** into the first pressure control chamber **1122** by the configuration to satisfy the relation $P1\geq P2$. As a consequence, the pressure P2 in the first pressure control chamber **1122** is not reduced further and the pressure P2 is maintained at the pressure within the predetermined range.

(127) On the other hand, in the case where the closed state of the communication port **1191** is established by a state of non-contact between the pressing plate **1210** and the valve shaft **1190***a* as shown in FIG. **17**C, a relation of the forces that act on the pressing plate **1210** is expressed by the following formula 3:

$$P3\times S3+F3=0 \qquad \text{Formula 3.}$$

(128) Here, rearrangement of the formula 3 with respect to P3 gives the following formula:

$$P3=-F3/S3 \qquad \text{Formula 4,}$$

where F3: a pressure adjustment spring **1220** in the state of non-contact between the pressing plate **1210** and the valve shaft **1190***a;* P3: a pressure (a gauge pressure) in the first pressure control chamber **1122** in the state of non-contact between the pressing plate **1210** and the valve shaft **1190***a*; and S**3**: a pressure receiving area of the pressing plate **1210** in the state of non-contact between the pressing plate **1210** and the valve shaft **1190***a.*

(129) Here, FIG. **17**C represents a state where the pressing plate **1210** and the flexible member **1230** are displaced to a displaceable limit in the rightward direction in FIG. **17**C. The pressure P3 in the first pressure control chamber **1122**, the spring force F3 of the pressure adjustment spring **1220**, and the pressure receiving area S**3** of the pressing plate **1210** vary in accordance with amounts of displacement of the pressing plate **1210** and the flexible member **1230** in the course of the displacement to the state in FIG. **17**C. To be more precise, in the case where the pressing plate **1210** and the flexible member **1230** are located in the leftward direction of FIG. **17**C as compared to the state illustrated in FIG. **17**C, the pressure receiving area S**3** of the pressing plate **1210** is reduced and the spring force F3 of the pressure adjustment spring **1220** is increased. As a consequence, the pressure P3 in the first pressure control chamber **1122** is reduced due to the relation of the formula 4. Accordingly, due to the formula 2 and the formula 4, the pressure in the first pressure control chamber **1122** is gradually increased (that is, the negative pressure is reduced to a value close to a positive pressure side) during a period of transition from the state in FIG. **17**B

to the state in FIG. **17**C. In other words, the pressure in the first pressure control chamber **1122** is gradually increased during the period in which the pressing plate **1210** and the flexible member **1230** are gradually displaced in the rightward direction from the state where the communication port **1191** is in the open state and the inner volume of the first pressure control chamber **1122** eventually reaches the displaceable limit. In short, the negative pressure is gradually reduced.

Circulation Pump

(130) Next, a configuration and operation of a circulation pump **1500** that is built in the above-described liquid ejection head **1000** will be described in more detail with reference to FIGS. **18**A, **18**B, and **19**.

(131) FIGS. **18**A and **18**B are external perspective views of the circulation pump **1500**. FIG. **18**A is the external perspective view showing a front side of the circulation pump **1500**, and FIG. **18**B is the external perspective view showing a back side of the circulation pump **1500**. An outer shell of the circulation pump **1500** is formed from a pump housing **1505** and a cover **1507** that is fixed to the pump housing **1505**. The pump housing **1505** is formed from a housing body **1505***a* and a flow passage connection member **1505***b* that is attached and fixed to an outer surface of the housing body **1505***a*. The housing body **1505***a* and the flow passage connection member **1505***b* are each provided with a pair of through holes at two different positions. The through holes formed at each position communicate with each other. The pair of through holes provided at one position collectively form a pump supply hole **1501** while the pair of through holes provided at another position collectively form a pump discharge hole **1502**. The pump supply hole **1501** is connected to a pump inlet flow passage **1170** that is connected to the second pressure control chamber **1152**, and the pump discharge hole **1502** is connected to a pump outlet flow passage **1180** that is connected to the first pressure control chamber **1122**. The ink supplied from the pump supply hole **1501** passes through a pump chamber **1503** (see FIG. **19**) to be described later, and is discharged from the pump discharge hole **1502**.

(132) FIG. **19** is a cross-sectional view of the circulation pump **1500** shown in FIG. **18**A, which is taken along the XIX-XIX line therein. A diaphragm **1506** is bonded to an inner surface of the pump housing **1505**, and the pump chamber **1503** is formed between this diaphragm **1506** and a recess formed in the inner surface of the pump housing **1505**. The pump chamber **1503** communicates with the pump supply hole **1501** and the pump discharge hole **1502**, which are formed in the pump housing **1505**. Meanwhile, a check valve **1504***a* is provided at an intermediate portion of the pump supply hole **1501**, and a check valve **1504***b* is provided at an intermediate portion of the pump discharge hole **1502**. To be more precise, the check valve **1504***a* is disposed such that part of the check valve **1504***a* can move leftward in FIG. **19** in a space **1512***a* defined at the intermediate portion of the pump supply hole **1501**. Meanwhile, the check valve **1504***b* is disposed such that part of the check valve **1504***b* can move rightward in FIG. **19** in a space **1512***b* defined at the intermediate portion of the pump discharge hole **1502**.

(133) In the case where the pressure in the pump chamber **1503** is reduced by an increase in volume of the pump chamber **1503** along with displacement of the diaphragm **1506**, the check valve **1504***a* is detached (that is, moves leftward in FIG. **19**) from an opening of the pump supply hole **1501** in the space **1512***a*. The detachment of the check valve **1504***a* from the opening of the pump supply hole **1501** in the space **1512***a* establishes the open state that enables the flow of the ink through the pump supply hole **1501**. On the other hand, in the case where the pressure in the pump chamber **1503** is increased by a decrease in volume of the pump chamber **1503** along with the displacement of the diaphragm **1506**, the check valve **1504***a* comes into close contact with a wall surface around the opening of the pump supply hole **1501**. As a consequence, the closed state is established to block the flow of the ink through the pump supply hole **1501**.

(134) Meanwhile, in the case where the pressure in the pump chamber **1503** is reduced, the check valve **1504***b* comes into close contact with a wall surface around an opening of the pump housing **1505**, and the closed state is established to block a flow of the ink through the pump discharge hole

1502. On the other hand, in the case where the pressure in the pump chamber **1503** is increased, the check valve **1504***b* is detached (that is, moves rightward in FIG. **19**) from the opening of the pump housing **1505** and moves toward the space **1512***b*, thus enabling the flow of the ink through the pump discharge hole **1502**.
(135) Here, a material of the respective check valves **1504***a* and **1504***b* only needs to have a deformable property in accordance with the pressure inside the pump chamber **1503**. The check valves can be formed from an elastic member such as EPDM and an elastomer, or from a film or a thin plate of polypropylene and the like. However, the applicable materials are not limited only to these materials.
(136) As described above, the pump chamber **1503** is formed by bonding the pump housing **1505** to the diaphragm **1506**. Accordingly, the pressure in the pump chamber **1503** varies with the deformation of the diaphragm **1506**. For example, the pressure in the pump chamber **1503** is increased in the case where the volume of the pump chamber **1503** is reduced by the displacement of the diaphragm **1506** toward the pump housing **1505** (rightward displacement in FIG. **19**). Hence, the check valve **1504***b* disposed opposite to the pump discharge hole **1502** is set to the open state and the ink in the pump chamber **1503** is discharged. In this instance, the check valve **1504***a* disposed opposite to the pump supply hole **1501** comes into close contact with the wall surface around the pump supply hole **1501**. Accordingly, a reverse flow of the ink from the pump chamber **1503** to the pump supply hole **1501** is suppressed.
(137) On the other hand, the pressure in the pump chamber **1503** is reduced in the case where the diaphragm **1506** is displaced in the direction to expand the pump chamber **1503**. Hence, the check valve **1504***a* disposed opposite to the pump supply hole **1501** is set to the open state and the ink is supplied to the pump chamber **1503**. In this instance, the check valve **1504***b* disposed at the pump discharge hole **1502** comes into close contact with the wall surface around the opening formed in the pump housing **1505** and occludes the opening. Accordingly, a reverse flow of the ink from the pump discharge hole **1502** to the pump chamber **1503** is suppressed.
(138) As described above, in the circulation pump **1500**, the ink is suctioned and discharged by changing the pressure in the pump chamber **1503** with the displacement of the diaphragm **1506**. In a case where the bubbles enter the pump chamber **1503** in this instance, the change in pressure in the pump chamber **1503** is diminished by expansion and contraction of the bubbles in spite of the displacement of the diaphragm **1506**, whereby a liquid feeding amount is reduced. Therefore, the pump chamber **1503** is disposed parallel to the gravitational force so as to let the bubbles entering the pump chamber **1503** gather easily at an upper part of the pump chamber **1503**, and the pump discharge hole **1502** is disposed at a portion above the center of the pump chamber **1503**. In this way, it is possible to improve a performance to discharge the bubbles in the pump and to stabilize a flow volume.
Ink Flow in Liquid Ejection Head
(139) FIGS. **20**A to **20**E are diagrams for explaining the flow of the ink in the liquid ejection head. Circulation of the ink to be carried out in the liquid ejection head **1000** will be described with reference to FIGS. **20**A to **20**E. In order to explain the circulation path for the ink more clearly, relative positions of the respective structures (the first pressure adjustment unit **1120**, the second pressure adjustment unit **1150**, the circulation pump **1500**, and so forth) are simplified in FIGS. **20**A to **20**E. For this reason, the relative positions of the structures are different from those of structures in FIG. **28** to be described later. FIG. **20**A schematically shows the flow of the ink in a case of carrying out a printing operation to perform printing while ejecting the ink from an ejection port **1013**. Note that arrows in FIG. **20**A indicate the flow of the ink. In the present embodiment, both an external pump **1021** and the circulation pump **1500** start driving in the case of carrying out the printing operation. Here, the external pump **1021** and the circulation pump **1500** may be driven irrespective of the printing operation. Alternatively, the external pump **1021** and the circulation pump **1500** do not have to be driven in tandem and may be driven independently of each other.

(140) The circulation pump **1500** is in on-state (a driven state) during the printing operation, and the ink flowing out of the first pressure control chamber **1122** flows into a supply flow passage **1130** and a bypass flow passage **1160**. The ink that flows into the supply flow passage **1130** passes through an ejection module **1300** and then flows into a collection flow passage **1140**. Thereafter, the ink is supplied to the second pressure control chamber **1152**.

(141) In the meantime, the ink flowing from the first pressure control chamber **1122** into the bypass flow passage **1160** passes through the second valve chamber **1151** and flows into the second pressure control chamber **1152**. The ink flowing into the second pressure control chamber **1152** passes through the pump inlet flow passage **1170**, the circulation pump **1500**, and the pump outlet flow passage **1180**, and then flows into the first pressure control chamber **1122** again. In this instance, a control pressure by the first valve chamber **1121** is set higher than a control pressure of the first pressure control chamber **1122** based on the relation of the above-mentioned formula 2. Accordingly, the ink in the first pressure control chamber **1122** is supplied to the ejection module **1300** through the supply flow passage **1130** again without flowing into the first valve chamber **1121**. The ink flowing into the ejection module **1300** passes through the collection flow passage **1140**, the second pressure control chamber **1152**, the pump inlet flow passage **1170**, the circulation pump **1500**, and the pump outlet flow passage **1180**, and then flows into the first pressure control chamber **1122** again. Thus, the ink circulation is carried out as described above, which is completed inside the liquid ejection head **1000**.

(142) In the above-described ink circulation, an amount of circulation (the flow volume) of the ink in the ejection module **1300** is determined by a differential pressure between the control pressures of the first pressure control chamber **1122** and the second pressure control chamber **1152**. Then, this differential pressure is set to achieve such an amount of circulation that can suppress an increase in viscosity of the ink in the vicinity of each ejection port in the ejection module **1300**. Moreover, the ink in an amount equivalent to an amount consumed by the printing is supplied from the ink tank **2** to the first pressure control chamber **1122** through a filter **1110** and the first valve chamber **1121**. A mechanism to supply the ink in the amount of consumption will be described below in detail. The pressure inside the first pressure control chamber is reduced in accordance with the decrease of the ink in the circulation path in the amount equivalent to the amount of the ink consumed by the printing. As a consequence, the ink in the first pressure control chamber **1122** is decreased as well. Along with the decrease of the ink in the first pressure control chamber **1122**, the inner volume of the first pressure control chamber **1122** is reduced. Due to this reduction in inner volume of the first pressure control chamber **1122**, a communication port **1191**A is set to the open state and the ink is supplied from the first valve chamber **1121** to the first pressure control chamber **1122**. A pressure loss occurs in this supplied ink in the process of passage from the first valve chamber **1121** through the communication port **1191**A, and the ink at a positive pressure is turned into a state of a negative pressure as a consequence of flowing into the first pressure control chamber **1122**. Then, the flow of the ink from the first valve chamber **1121** into the first pressure control chamber **1122** brings about a rise in pressure inside the first pressure control chamber, thereby increasing the inner volume of the first pressure control chamber and establishing the closed state of the communication port **1191**A. In this way, the open state and the closed state are repeated in the communication port **1191**A along with the ink consumption. In the meantime, the communication port **1191**A is maintained in the closed state in the case where the ink is not consumed.

(143) FIG. **20**B schematically shows the flow of the ink immediately after the printing operation is completed and the circulation pump **1500** transitions to off state (a stopped state). At the point of completion of the printing operation and in the off state of the circulation pump **1500**, both the pressure in the first pressure control chamber **1122** and the pressure in the second pressure control chamber **1152** are set to the control pressures in the course of the printing operation. For this reason, movement of the ink is generated as shown in FIG. **20**B in accordance with the differential

pressure between the pressure in the first pressure control chamber **1122** and the pressure in the second pressure control chamber **1152**. To be more precise, the flow of the ink to be supplied from the first pressure control chamber **1122** to the ejection module **1300** through the supply flow passage **1130** and then to reach the second pressure control chamber **1152** through the collection flow passage **1140** is continuously generated. Meanwhile, the flow of the ink from the first pressure control chamber **1122** to the second pressure control chamber **1152** through the bypass flow passage **1160** and the second valve chamber **1151** is continuously generated as well.
(144) Due to these flows of the ink, the ink in an amount equivalent to the amount of ink having moved from the first pressure control chamber **1122** to the second pressure control chamber **1152** is supplied from the ink tank **2** to the first pressure control chamber **1122** through the filter **1110** and the first valve chamber **1121**. As a consequence, an internal content in the first pressure control chamber **1122** is kept constant. Based on the relation of the above-mentioned formula 2, the spring force F1 of the valve spring **1200**, the spring force F2 of the pressure adjustment spring **1220**, the pressure receiving area Si of the valve **1190**, and the pressure receiving area S2 of the pressing plate **1210** are kept constant in the case where the internal content in the first pressure control chamber **1122** is constant. Accordingly, the pressure in the first pressure control chamber **1122** is determined in accordance with the change in pressure (gauge pressure) P1 in the first valve chamber **1121**. Therefore, in the case where there is no change of the pressure P1 in the first valve chamber **1121**, the pressure P2 in the first pressure control chamber **1122** is maintained at the same pressure as the control pressure during the printing operation.
(145) On the other hand, the pressure in the second pressure control chamber **1152** varies over time in accordance with a change in internal content associated with the inflow of the ink from the first pressure control chamber **1122**. To be more precise, during a period of transition from the state in FIG. **20**B to a state where the state of non-communication between the second valve chamber **1151** and the second pressure control chamber **1152** takes place due to establishment of the closed state of the communication port **1191** as shown in FIG. **20**C, the pressure in the second pressure control chamber **1152** varies in accordance with the formula 2. Thereafter, the pressing plate **1210** and the valve shaft **1190***a* transition to the state of non-contact whereby the closed state of the communication port **1191** is established. Then, the ink flows from the collection flow passage **1140** into the second pressure control chamber **1152** as shown in FIG. **20**D. Due to this ink inflow, the pressing plate **1210** and the flexible member **1230** are displaced. Hence, the pressure in the second pressure control chamber **1152** varies, or more specifically, increases in accordance with the formula 4 until the internal content in the second pressure control chamber **1152** reaches a maximum.
(146) Note that the flow of the ink from the first pressure control chamber **1122** to the second pressure control chamber **1152** through the bypass flow passage **1160** and the second valve chamber **1151** is not generated in the case where the state shown in FIG. **20**C takes place. Accordingly, there is only generated the flow of the ink in the first pressure control chamber **1122** to be supplied to the ejection module **1300** through the supply flow passage **1130** and then to reach the second pressure control chamber **1152** through the collection flow passage **1140**. As described above, the movement of the ink from the first pressure control chamber **1122** to the second pressure control chamber **1152** is generated in accordance with the differential pressure between the pressure in the first pressure control chamber **1122** and the pressure in the second pressure control chamber **1152**. For this reason, the movement of the ink is stopped in the case where the pressure in the second pressure control chamber **1152** becomes equal to the pressure in the first pressure control chamber **1122**.
(147) Meanwhile, in the state where the pressure in the second pressure control chamber **1152** is equal to the pressure in the first pressure control chamber **1122**, the second pressure control chamber **1152** expands to a state shown in FIG. **20**D. In the case where the second pressure control chamber **1152** expands as shown in FIG. **20**D, a reservoir unit capable of reserving the ink is

formed in the second pressure control chamber **1152**. Although it may vary depending on the shape and the size of the flow passage and on properties of the ink, the stopped state of the circulation pump **1500** transitions to the state shown in FIG. **20**D within a period of about 1 to 2 minutes. In the case of driving the circulation pump **1500** in the state of reserving the ink in the reservoir unit as shown in FIG. **20**D, the ink in the reservoir unit is supplied to the first pressure control chamber **1122** by using the circulation pump **1500**. Thus, the amount of the ink in the first pressure control chamber **1122** is increased as shown in FIG. **20**E, and the flexible member **1230** and the pressing plate **1210** are displaced in an expanding direction. Then, as the circulation pump **1500** is continuously driven, the state inside the circulation path will be changed as shown in FIG. **20**A.
(148) In the above description, FIG. **20**A has been explained as the example in the case of the printing operation. However, as mentioned earlier, the ink may be circulated irrespective of execution of the printing operation. In this case as well, the flow of the ink is generated as shown in FIGS. **20**A to **20**E in accordance with the drive and the stop of the circulation pump **1500**.
(149) As described above, the present embodiment adopts the example in which a communication port **1191**B in the second pressure adjustment unit **1150** is set to the open state in the case where the ink is circulated by driving the circulation pump **1500**, and is set to the closed state in the case where the circulation of the ink is stopped. However, the present disclosure is not limited only to this configuration. The control pressure in the communication port **1191**B in the second pressure adjustment unit **1150** may be set so as to establish the closed state even in the case where the ink is circulated by driving the circulation pump **1500**. This configuration will be specifically described below together with a function of the bypass flow passage **1160**.
(150) The bypass flow passage **1160** to connect the first pressure adjustment unit **1120** to the second pressure adjustment unit **1150** is provided in order not to adversely affect the ejection module **1300** in a case where the negative pressure generated in the circulation path exceeds a predetermined value, for example. Moreover, the bypass flow passage **1160** is also provided in order to supply the ink from both the supply flow passage **1130** side and the collection flow passage **1140** side into a pressure chamber **1012**.
(151) The example of providing the bypass flow passage **1160** in order not to adversely affect the ejection module **1300** in the case where the negative pressure exceeds the predetermined value will be described to begin with. A property (such as viscosity) of the ink may be altered by a change in environmental temperature, for instance. In the case where the viscosity of the ink is altered, the pressure loss in the circulation path is changed as well. For example, the pressure loss in the circulation path is reduced in the case where the viscosity of the ink is decreased. As a consequence, a flow rate of the circulation pump **1500** being driven at a constant drive amount is increased and the volume of the flow in the ejection module **1300** is increased as a consequence. On the other hand, the ejection module **1300** is maintained at a constant temperature by using a not-illustrated temperature adjustment mechanism. Accordingly, the viscosity of the ink in the ejection module **1300** is kept constant even in the case of a change in environmental temperature. Since the flow rate of the ink flowing in the ejection module **1300** is increased while there is no change in viscosity of the ink in the ejection module **1300**, the negative pressure in the ejection module **1300** is increased due to flow resistance. In the case where the negative pressure in the ejection module **1300** exceeds the predetermined value as escribed above, a meniscus on the ejection port **1013** may be destroyed and normal ejection may therefore be infeasible. Even in a case where the meniscus is spared from destruction, the negative pressure in the pressure chamber **1012** exceeds a prescribed value and ejection therefrom may be adversely affected.
(152) Given the circumstances, the bypass flow passage **1160** is formed inside the circulation path in the present embodiment. By providing the bypass flow passage **1160**, the ink also flows in the bypass flow passage **1160** in the case where the negative pressure exceeds the predetermined value. Thus, the pressure in the ejection module **1300** can be kept constant. Accordingly, the communication port **1191**B in the second pressure adjustment unit **1150** may be provided with such

a control pressure that maintains the closed state even in the case where the circulation pump **1500** is being driven. Moreover, the control pressure in the second pressure adjustment unit **1150** may be set such that the communication port **1191** in the second pressure adjustment unit **1150** establishes the open state in the case where the negative pressure exceeds the predetermined value. In other words, the communication port **1191**B may be in the closed state in the case of driving the circulation pump **1500** as long as the meniscus is not destroyed by the change in flow rate in the pump due to the change in viscosity such as an environmental change, or as long as the predetermined negative pressure is maintained.

Configuration of Ejection Unit

(153) FIGS. **21**A and **21**B are schematic diagrams showing a circulation path for one ink color in an ejection unit **1003** of the present embodiment. FIG. **21**A is an exploded perspective view of the ejection unit **1003** viewed from a first support member **1004** side, and FIG. **21**B is an exploded perspective view of the ejection unit **1003** viewed from the ejection module **1300** side. Note that each of arrows with IN and OUT remarks indicated in the FIGS. **21**A and **21**B shows a flow of the ink. Although the flows of the ink for one color will be discussed herein, inks of other colors exhibit similar flows. Meanwhile, illustration of a second support member and electric wiring members is omitted in FIGS. **21**A and **21**B. Explanations of these constituents are also omitted in the following description of the configuration of the ejection unit. The ejection module **1300** includes an ejection element board **1340** and an opening plate **1330**. FIG. **22** is a diagram showing the opening plate **1330**, and FIG. **23** is a diagram showing the ejection element board **1340**.

(154) The ink is supplied from the circulation unit **200** to the ejection unit **1003** through a not-illustrated joint member. A description will be given of the ink path from a point after the ink passes through the joint member to a point of return of the ink to the joint member.

(155) The ejection module **1300** includes the ejection element board **1340**, which is a silicon substrate **1310**, and the opening plate **1330**. The ejection module **1300** also includes an ejection port forming member **1320**. The ejection element board **1340**, the opening plate **1330**, and the ejection port forming member **1320** are stacked on and bonded to one another in such a way as to establish communication of the flow passages for the respective inks, thus constituting the ejection module **1300** that is supported by the first support member **1004**. The ejection unit **1003** is formed by supporting the ejection module **1300** by the first support member **1004**. The ejection element board **1340** includes the ejection port forming member **1320**. The ejection port forming member **1320** includes ejection port rows each formed from the ejection ports **1013** arranged in a row. Part of the ink supplied through ink flow passages in the ejection module **1300** is ejected from each ejection port **1013**. The ink which is not ejected is recovered through the ink flow passages in the ejection module **1300**.

(156) As shown in FIGS. **21**A, **21**B, and **22**, the opening plate **1330** includes arranged ink supply ports **1311** and arranged ink collection ports **1312**. As shown in FIGS. **23** and **24**A to **24**C, the ejection element board **1340** includes arranged supply connection flow passages **1323** and arranged collection connection flow passages **1324**. In addition, the ejection element board **1340** includes common supply flow passages **1018** that communicate with the supply connection flow passages **1323**, and common collection flow passages **1019** that communicate with the collection connection flow passages **1324**. The ink flow passages in the ejection unit **1003** are formed by causing ink supply flow passages **1048** and ink collection flow passages **1049**, which are provided to the first support member **1004**, to communicate with the flow passages that are provided to the ejection module **1300**. Support member supply ports **1211** are cross-sectional openings that constitute the ink supply flow passages **1048** and support member collection ports **1212** are cross-sectional openings that constitute the ink collection flow passages **1049**.

(157) The ink to be supplied to the ejection unit **1003** is supplied from the circulation unit **200** side to the ink supply flow passages **1048** of the first support member **1004**. The ink flowing through the support member supply ports **1211** in the ink supply flow passages **1048** is supplied to the

common supply flow passages **1018** of the ejection element board **1340** through the ink supply flow passages **1048** and the ink supply ports **1311** of the opening plate **1330**, and then enters the supply connection flow passages **1323**. These flow passages collectively constitute supply side flow passages. Thereafter, the ink flows to the collection connection flow passages **1324** of collection side flow passages through the pressure chambers **1012** of the ejection port forming member **1320**. Details of the flow of the ink in each pressure chamber **1012** will be described later.

(158) In the collection side flow passages, the ink entering the collection connection flow passages **1324** flows in the common collection flow passages **1019**. Thereafter, the ink flows from the common collection flow passages **1019** to the ink collection flow passages **1049** of the first support member **1004** through the ink collection ports **1312** of the opening plate **1330**, and is recovered by the circulation unit **200**.

(159) A region in the opening plate **1330** not provided with the ink supply ports **1311** or the ink collection ports **1312** corresponds to a region for partitioning the support member supply ports **1211** and the support member collection ports **1212** in the first support member **1004**. Moreover, no openings are provided to the first support member **1004** in this region. The above-mentioned region is used as an attachment region in a case of attaching the ejection module **1300** to the first support member **1004**.

(160) In the opening plate **1330** in FIG. **22**, rows of openings arranged in the x direction are provided in the y direction. Here, supply (IN) openings and collection (OUT) openings are alternately arranged in the y direction in such a way as to be displaced by a half pitch in the x direction. In the ejection element board **1340** in FIG. **23**, the common supply flow passages **1018** communicating with the supply connection flow passages **1323** arranged in the y direction and the common collection flow passages **1019** communicating with the collection connection flow passages **1324** arranged in the y direction are alternately arranged in the x direction. The common supply flow passages **1018** and the common collection flow passages **1019** are separated by the types of the inks. Moreover, the numbers of the common supply flow passages **1018** and the common collection flow passages **1019** disposed are determined in accordance with the numbers of ejection port rows of the respective colors. In the meantime, the supply connection flow passages **1323** and the collection connection flow passages **1324** are also arranged in the numbers corresponding to the ejection ports **1013**. Here, the supply connection flow passages **1323** and the collection connection flow passages **1324** do not always have to correspond one-to-one to the ejection ports **1013**. One supply connection flow passage **1323** and one collection connection flow passage **1324** may deal with two or more ejection ports **1013**.

(161) The opening plate **1330** and the ejection element board **1340** described above are stacked on and bonded to each other such that the flow passages for the respective inks establish communication, thus being formed into the ejection module **1300** which is supported by the first support member **1004**. In this way, the ink flow passages are formed which include the supply flow passages and the collection flow passages as described above.

(162) FIGS. **24**A to **24**C are cross-sectional views showing the flow of the ink at different portions of the ejection unit **1003**. FIG. **24**A shows a cross-sectional view taken along the XXIVA-XXIVA line in FIG. **21**A, which represents a cross-section of a portion where the ink supply flow passages **1048** communicate with the ink supply ports **1311** in the ejection unit **1003**. Meanwhile, FIG. **24**B shows a cross-sectional view taken along the XXIVB-XXIVB line in FIG. **21**A, which represents a cross-section of a portion where the ink collection flow passages **1049** communicate with the ink collection ports **1312** in the ejection unit **1003**. In the meantime, FIG. **24**C shows a cross-sectional view taken along the XXIVC-XXIVC line in FIG. **21**A, which represents a cross-section of a portion where the ink supply ports **1311** or the ink collection ports **1312** do not communicate with the flow passages of the first support member **1004**.

(163) In the supply flow passages to supply the ink, the ink is supplied from portions where the ink supply flow passages **1048** of the first support member **1004** overlap and communicate with the ink

supply ports **1311** of the opening plate **1330** as shown in FIG. **24**A. Meanwhile, in the collection flow passages to recover the ink, the ink is recovered from portions where the ink collection flow passages **1049** of the first support member **1004** overlap and communicate with the ink collection ports **1312** of the opening plate **1330** as shown in FIG. **24**B. In the meantime, there are also regions in the ejection unit **1003** where openings are not partially provided to the opening plate **1330**. In such a region, the ink is not supplied or recovered between the ejection element board **1340** and the first support member **1004**. The ink is supplied in the regions where the ink supply ports **1311** are provided as shown in FIG. **24**A, and the ink is recovered in the regions where the ink collection ports **1312** are provided as shown in FIG. **24**B. Although the present embodiment has described an example of the configuration adopting the opening plate **1330**, a mode of not using the opening plate **1330** is also acceptable. For instance, a configuration to provide the first support member **1004** with the flow passages corresponding to the ink supply flow passages **1048** and the ink collection flow passages **1049**, and to bond the ejection element board **1340** to the first support member **1004** is also acceptable.

(164) FIGS. **25**A and **25**B are cross-sectional views showing a portion in the vicinity of a certain ejection port **1013** in the ejection module **1300**. Note that thick arrows indicated in a common supply flow passage **1018** and a common collection flow passage **1019** in FIGS. **25**A and **25**B represent swing of the ink in a mode of using the serial type liquid ejection apparatus **2000**. The ink supplied to the pressure chamber **1012** through the common supply flow passage **1018** and the supply connection flow passage **1323** is ejected from the ejection port **1013** by driving an ejection element **1015**. In the case where the ejection element **1015** is not driven, the ink passes through the pressure chamber **1012** and the collection connection flow passage **1324** being the collection flow passage, and is recovered by the common collection flow passage **1019**.

(165) In the case of ejecting the circulating ink in the mode of using the serial type liquid ejection apparatus **2000** as described above, the ejection of the ink is affected more than a little by the swing of the ink in the ink flow passage, which is attributed to main scanning of the liquid ejection head **1000**. To be more precise, an impact of the swing of the ink in the ink flow passage manifests as a difference in amount of ejection of the ink or as deviation in direction of ejection.

(166) Given the circumstances, the common supply flow passage **1018** and the common collection flow passage **1019** of the present embodiment is configured to extend in the y direction in the cross-sections shown in FIGS. **25**A and **25**B, and also to extend in the z direction that is perpendicular to the x direction being the main scanning direction. This configuration can reduce the width in the main scanning direction of each of the common supply flow passage **1018** and the common collection flow passage **1019**. The swing of the ink attributable to a force of inertial (black thick arrows in FIGS. **25**A and **25**B to be applied in an opposite direction from the scanning direction, which acts on the ink in the common supply flow passage **1018** and the common collection flow passage **1019** in the course of main scanning, is diminished by reducing the width in the main scanning direction of each of the common supply flow passage **1018** and the common collection flow passage **1019**. In this way, it is possible to suppress an adverse effect of the swing of the ink on the ink ejection. Moreover, each of the common supply flow passage **1018** and the common collection flow passage **1019** extends in the z direction in order to increase the cross-sectional area, thereby reducing pressure drops in the flow passages.

(167) As described above, the common supply flow passage **1018** and the common collection flow passage **1019** are configured to reduce the swing of the ink therein by setting the small widths of the common supply flow passage **1018** and the common collection flow passage **1019** in the main scanning direction. Nevertheless, this configuration cannot completely eliminate the swing. Therefore, the present embodiment is configured to deploy the common supply flow passages **1018** and the common collection flow passages **1019** at positions overlapping one another in the x direction so as to suppress the occurrence of a difference in ejection among the inks of the different types that may develop even by the reduced amount of swing.

(168) As described above, in the present embodiment, the supply connection flow passages **1323** and the collection connection flow passages **1324** are provided corresponding to the ejection ports **1013**. Moreover, the supply connection flow passages **1323** and the collection connection flow passages **1324** have such a correspondence relation to be juxtaposed in the x direction while interposing the ejection ports **1013** in between. Accordingly, there are portions where the common supply flow passages **1018** do not overlap the common collection flow passages **1019** in the x direction. In a case where the correspondence relation in the x direction between the supply connection flow passages **1323** and the collection connection flow passages **1324** breaks up, the flow and ejection of the ink in the x direction in the pressure chambers **1012** may be adversely affected. Here, addition of the adverse effect of the swing of the ink may have a larger impact on ejection of the ink from each ejection port.
(169) For this reason, the common supply flow passages **1018** are disposed at such positions overlapping the common collection flow passages **1019** in the x direction. In this way, the swing of the ink in the common supply flow passage **1018** at the time of the main scanning is substantially equal to the swing of the ink in the corresponding common collection flow passage **1019** at any position in the y direction in which the ejection ports **1013** are arranged. As a consequence, it is possible to achieve stable ejection while avoiding a significant variation in pressure difference between the common supply flow passage **1018** side and the common collection flow passage **1019** side, which may take place in each pressure chamber **1012**.
(170) Meanwhile, some liquid ejection heads that circulate the ink may be configured to use the same flow passage for forming the flow passage to supply the ink to the liquid ejection head and the flow passage to recover the ink therefrom. On the other hand, according to the present embodiment, the common supply flow passage **1018** and the common collection flow passage **1019** are provided as separate flow passages. Moreover, each pressure chamber **1012** communicates with the supply connection flow passage **1323** and the pressure chamber **1012** also communicates with the collection connection flow passage **1324**. Hence, the ink is ejected from the ejection port **1013** of the pressure chamber **1012**. In other words, the pressure chamber **1012** serving as the path to join the supply connection flow passage **1323** to the collection connection flow passage **1324** is also provided with the ejection port **1013**. Accordingly, the flow of the ink that flows from the supply connection flow passage **1323** side to the collection connection flow passage **1324** side is generated in the pressure chamber **1012**, and the ink in the pressure chamber **1012** is efficiently circulated. The efficient circulation of the ink in the pressure chamber **1012** can keep the ink in the pressure chamber **1012** in a fresh condition although this ink is susceptible to evaporation from the ejection port **1013**.
(171) In the meantime, the two flow passages of the common supply flow passage **1018** and the common collection flow passage **1019** communicate with the corresponding pressure chamber **1012**. Accordingly, it is also possible to supply the ink from both of the flow passages in the case where ejection has to be carried out at a high flow rate. In other words, as compared to the configuration to supply and recover the ink with only one flow passage, the configuration of the present embodiment has an advantage that it is possible not only to perform the circulation efficiently but also to deal with ejection at a high flow rate.
(172) Meanwhile, the adverse effect of the swing of the ink becomes even less in the case where the common supply flow passage **1018** and the common collection flow passage **1019** are located at positions close to each other in the x direction. Such an interval between the flow passages may desirably be set in a range from 75 to 100 μm.
(173) FIG. **26** is a diagram showing an ejection element board **1340** of a comparative example. Note that illustration of the supply connection flow passages **1323** and the collection connection flow passages **1324** is omitted in FIG. **26**. The ink that receives thermal energy from the ejection element **1015** in the pressure chamber **1012** flows into the common collection flow passage **1019**. Accordingly, the ink flowing therein has a relatively higher temperature than a temperature of the

ink in the common supply flow passage **1018**. In this instance, in the comparative example, there is a portion in the x direction of the ejection element board **1340**, such as a portion a surrounded by a chain line in FIG. **26**, which is a portion where only the common collection flow passages **1019** are present. In this case, the temperature is locally increased in the relevant portion. Accordingly, a temperature variation may occur in the ejection module **1300** and may adversely affect the ejection.
(174) The ink flowing in the common supply flow passage **1018** has a relatively low temperature as compared to that in the common collection flow passage **1019**. For this reason, in the case where the common supply flow passage **1018** and the common collection flow passage **1019** are located adjacent to each other, a certain degree of the temperatures in the common supply flow passage **1018** and in the common collection flow passage **1019** are cancelled and a rise in temperature is therefore suppressed. Accordingly, the common supply flow passage **1018** and the common collection flow passage **1019** having substantially the same lengths are preferably present at positions overlapping each other in the x direction and adjacent to each other.
(175) FIGS. **27**A and **27**B are diagrams showing a flow passage configuration of the liquid ejection head **1000** adaptable to inks of three colors of cyan (C), magenta (M), and yellow (Y). As shown in FIG. **27**A, the liquid ejection head **1000** is provided with circulation flow passages for the respective ink types. The pressure chambers **1012** are provided in the x direction being the main scanning direction of the liquid ejection head **1000**. Meanwhile, as shown in FIG. **27**B, the common supply flow passage **1018** and the common collection flow passage **1019** are provided along the ejection port row on which the ejection port **1013** are arranged, the common supply flow passage **1018** and the common collection flow passage **1019** are provided extending in the y direction in such a way as to interpose the ejection port row in between.
Connection Between Body Unit and Liquid Ejection Head
(176) FIG. **28** is a schematic configuration diagram showing details of a state of connection of the liquid ejection head **1000** to the ink tank **2** and the external pump **1021** provided to a body unit of the liquid ejection apparatus **2000**, and a layout of the circulation pump and the like. The liquid ejection apparatus **2000** according to the present embodiment has such a configuration that facilitates replacement of the liquid ejection head **1000** only in a case of the occurrence of a failure in the liquid ejection head **1000**. To be more precise, there is provided a liquid connection unit **1700** that facilitates connection and detachment of the liquid ejection head **1000** to and from an ink supply tube **1059** that is connected to the external pump **1021**. This makes it possible to attach and detach only the liquid ejection head **1000** to and from the liquid ejection apparatus **2000** easily.
(177) As shown in FIG. **28**, the liquid connection unit **1700** includes a liquid connector insertion slot **1053***a* provided in a projecting manner to a head housing **1053** of the liquid ejection head **1000**, and a cylindrical liquid connector **1059***a* which can be inserted into this liquid connector insertion slot **1053***a*. The liquid connector insertion slot **1053***a* is fluidically connected to the ink supply flow passage formed inside the liquid ejection head **1000**, and is connected to the first pressure adjustment unit **1120** through the filter **1110** mentioned above. Meanwhile, the liquid connector **1059***a* is provided at a tip of the ink supply tube **1059** connected to the external pump **1021** that pressure-supplies the ink from the ink tank **2** to the liquid ejection head **1000**.
(178) As described above, the liquid ejection head **1000** shown in FIG. **28** facilitates attachment and detachment operations as well as a replacement operation of the liquid ejection head **1000** by using the liquid connection unit **1700**. However, the ink that is pressure-supplied by the external pump **1021** may leak out of the liquid connection unit **1700** in case of deterioration of a sealing performance between the liquid connector insertion slot **1053***a* and the liquid connector **1059***a*. In case the leaking ink adheres to the circulation pump **1500** and the like, a failure may occur in an electric system and the like. Given the circumstances, the circulation pump and the like are laid out as described below in the present embodiment.
Layout of Circulation Pump and Others
(179) As shown in FIG. **28**, in the present embodiment, the circulation pump **1500** is disposed

above the liquid connection unit **1700** in the gravitational direction in order to avoid adhesion of the ink to the circulation pump **1500** after leaking out of the liquid connection unit **1700**. Specifically, the circulation pump **1500** is disposed above the liquid connector insertion slot **1053***a* in the gravitational direction, the liquid connector insertion slot **1053***a* serving as the liquid inlet port of the liquid ejection head **1000**. Moreover, the circulation pump **1500** is located at a position so as not to come into contact with the components constituting the liquid connection unit **1700**. Accordingly, even in case the ink leaks out of the liquid connection unit **1700**, the ink will flow either in a horizontal direction being an opening direction of the liquid connector **1059***a* or downward in the gravitational direction. Thus, the ink can be kept from reaching the circulation pump **1500** located above in the gravitational direction. Moreover, since the circulation pump **1500** is located at the position distant from the liquid connection unit **1700**, it is unlikely that the ink reaches the circulation pump **1500** while flowing on other members.

(180) Meanwhile, an electric connection module **1515** for electrically connecting the circulation pump **1500** to an electric contact board **1006** through a flexible wiring member **1514** is provided above the liquid connection unit **1700** in the gravitational direction. This configuration can also reduce the chance of occurrence of electrical trouble due to the ink leaking out of the liquid connection unit **1700**.

(181) In the meantime, the head housing **1053** is provided with a wall portion **1053***b*. Accordingly, even in a case where the ink spouts from an opening **1059***b* of the liquid connection unit **1700**, it is possible to block the ink and to reduce the chance that the ink reaches the circulation pump **1500** or the electric connection module **1515**.

(182) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(183) This application claims the benefit of Japanese Patent Application No. 2022-081590 filed May 18, 2022, which is hereby incorporated by reference wherein in its entirety.

Claims

1. A liquid ejection apparatus comprising: a printing element board including a pressure chamber provided with an ejection port, wherein the printing element board is configured to eject a liquid from the ejection port; a supply flow passage provided to the printing element board and communicating with the pressure chamber; a collection flow passage provided to the printing element board and communicating with the pressure chamber; and a liquid feeding mechanism configured to generate a difference in pressure between the supply flow passage and the collection flow passage in such a way as to supply the liquid from the supply flow passage to the pressure chamber and to recover the liquid in the pressure chamber from the collection flow passage, and comprising a first bubble reservoir portion connecting the supply flow passage to the liquid feeding mechanism and a second bubble reservoir portion connecting the collection flow passage to the liquid feeding mechanism; wherein a volume of the first bubble reservoir portion is larger than a volume of the second bubble reservoir portion, wherein the first bubble reservoir portion includes a first pressure adjustment chamber capable of adjusting a pressure between the supply flow passage and the liquid feeding mechanism, and wherein the second bubble reservoir portion includes a second pressure adjustment chamber capable of adjusting a pressure between the collection flow passage and the liquid feeding mechanism.
2. The liquid ejection apparatus according to claim 1, wherein the printing element board, the supply flow passage, the collection flow passage, and the liquid feeding mechanism are provided to a liquid ejection head which is mounted on a carriage and is movable.
3. The liquid ejection apparatus according to claim 1, wherein the first bubble reservoir unit

includes a supply connection flow passage formed in a flow passage member to be stacked on the printing element board, and wherein the second bubble reservoir unit includes a collection connection flow passage formed in the flow passage member.
4. The liquid ejection apparatus according to claim 1, wherein the first pressure adjustment chamber and the second pressure adjustment chamber are provided to a circulation unit connected to a flow passage member stacked on the printing element board.
5. The liquid ejection apparatus according to claim 1, wherein a volume of the first pressure adjustment chamber is larger than a volume of the second pressure adjustment chamber.
6. The liquid ejection apparatus according to claim 1, wherein the first pressure adjustment chamber is connected to the second pressure adjustment chamber through the liquid feeding mechanism, and wherein a bypass flow passage configured to connect the first pressure adjustment chamber to the second pressure adjustment chamber without interposing the liquid feeding mechanism is provided between the first pressure adjustment chamber and the second pressure adjustment chamber.
7. The liquid ejection apparatus according to claim 1, wherein the first pressure adjustment chamber is connected to a liquid tank through a filter.
8. The liquid ejection apparatus according to claim 1, wherein the first bubble reservoir unit and the second bubble reservoir unit are each provided with a slit having a shape of a groove along a flow of the liquid flowing in the first bubble reservoir unit and the second bubble reservoir unit.
9. The liquid ejection apparatus according to claim 8, wherein a width of the slit is in a range from 0.2 to 0.5 mm.
10. The liquid ejection apparatus according to claim 1, wherein a flow passage to join the supply flow passage to the first pressure adjustment chamber extends in a vertical direction.
11. The liquid ejection apparatus according to claim 10, wherein a flow passage to join the supply flow passage to the first pressure adjustment chamber is inclined relative to a gravitational direction, and wherein the flow passage includes a flow passage inner wall where a component force of a normal vector has a component in the gravitational direction.
12. The liquid ejection apparatus according to claim 10, wherein the volume of the first bubble reservoir unit is at least 1.2 times as large as the volume of the second bubble reservoir unit.
13. A liquid ejection head comprising: a printing element board including a pressure chamber provided with an ejection port, wherein the printing element board is configured to eject a liquid from the ejection port; a supply flow passage provided to the printing element board and communicating with the pressure chamber; a collection flow passage provided to the printing element board and communicating with the pressure chamber; and a liquid feeding mechanism configured to generate a difference in pressure between the supply flow passage and the collection flow passage in such a way as to supply the liquid from the supply flow passage to the pressure chamber and to recover the liquid in the pressure chamber from the collection flow passage, and comprising a first bubble reservoir portion connecting the supply flow passage to the liquid feeding mechanism and a second bubble reservoir portion connecting the collection flow passage to the liquid feeding mechanism; wherein a volume of the first bubble reservoir portion is larger than a volume of the second bubble reservoir portion, wherein the first bubble reservoir portion includes a first pressure adjustment chamber capable of adjusting a pressure between the supply flow passage and the liquid feeding mechanism, and wherein the second bubble reservoir portion includes a second pressure adjustment chamber capable of adjusting a pressure between the collection flow passage and the liquid feeding mechanism.
14. The liquid ejection apparatus according to claim 13, wherein the printing element board, the supply flow passage, the collection flow passage, and the liquid feeding mechanism are provided to a liquid ejection head which is mounted on a carriage and is movable.
15. The liquid ejection apparatus according to claim 13, wherein the first bubble reservoir unit includes a supply connection flow passage formed in a flow passage member to be stacked on the

printing element board, and wherein the second bubble reservoir unit includes a collection connection flow passage formed in the flow passage member.

16. The liquid ejection apparatus according to claim 13, wherein the first pressure adjustment chamber and the second pressure adjustment chamber are provided to a circulation unit connected to a flow passage member stacked on the printing element board.

17. The liquid ejection apparatus according to claim 13, wherein a volume of the first pressure adjustment chamber is larger than a volume of the second pressure adjustment chamber.

18. The liquid ejection apparatus according to claim 13, wherein the first pressure adjustment chamber is connected to the second pressure adjustment chamber through the liquid feeding mechanism, and wherein a bypass flow passage configured to connect the first pressure adjustment chamber to the second pressure adjustment chamber without interposing the liquid feeding mechanism is provided between the first pressure adjustment chamber and the second pressure adjustment chamber.

19. The liquid ejection apparatus according to claim 13, wherein the first pressure adjustment chamber is connected to a liquid tank through a filter.

20. The liquid ejection apparatus according to claim 13, wherein the first bubble reservoir unit and the second bubble reservoir unit are each provided with a slit having a shape of a groove along a flow of the liquid flowing in the first bubble reservoir unit and the second bubble reservoir unit.